

Khyles Gibrian Ramos

Software Engineer · Web Developer

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GitHub: <https://github.com/NovaProtocol>

Portfolio: <portfolio.projectnova.download>



SUMMARY

DOST Scholar and engineering student with a foundation in software programming, system automation, and network configuration. Experience in scripting, web infrastructure, and software-hardware integration through academic and personal projects. Adaptable and eager to learn, with a desire to build practical, real-world skills in system architecture and backend operations.

CAREER OBJECTIVES

Motivated individual seeking an entry-level technical role to apply hands-on experience in coding, circuit prototyping, and system simulations. Eager to contribute to a collaborative team while actively learning from experienced professionals and continuously expanding my multidisciplinary skill set.

EDUCATION

Southern Luzon State University

2021-2026

Bachelor of Science in Mechanical Engineering

- DOST S&T Scholarship
- Dean's Lister

Sacred Heart College Lucena

2019-2021

Science Technology Engineering & Mathematics (STEM) Strand

- Academic Excellence Awardee - With Honours

Luis Palad Integrated High School

2015-2019

Junior High School

CERTIFICATION

DOLE Accredited Safety Officer 2 (SO2)

2024

Basic Occupational Safety and Health (BOSH SO2) Training

Completed 40-Hour OSH Course

WORK EXPERIENCE

Southern Luzon State University

Mar 2024 - Apr 2024

Physical Plant Facilities Department - PTP Program

- Conducted routine inventory audits and applied asset tags to university property to improve tracking accuracy.
- Inspected campus emergency lights and fire extinguishers to ensure proper working order and safety compliance.
- Utilized AutoCAD to design and update detailed facility layouts and emergency fire exit plans.
- Handled daily office operations, including documentation, message routing, and cross-department communications.

N/A - Work from Home

Aug 2025 - Nov 2025

Freelance Software Developer

- Developed a full-stack signature verification application using PyTorch, ResNet-18, and Flask to support academic thesis research.
- Built a signature verification system that utilizes Triplet Loss to compare known reference signatures against unknown samples for authentication.
- Developed a web-based user interface using Flask to allow users to upload signature samples and view real-time verification results.
- Integrated PyTorch models into the backend to handle real-time image processing and model inference upon user request.
- Created data preprocessing scripts to prepare, normalize, and format signature image datasets for model training.
- Delivered functional code and technical documentation within project deadlines.

PROJECTS

MELE Review

Tech Stack: FastAPI, Granian, SQLAlchemy, MySQL, Docker, Caddy, Cloudflare Tunnel

- Self-hosted board exam reviewer with a self-maintained question bank from various sources, with answer feedback and flag-for-review
- Notebook-style solution editor with a unit-aware solver that handles multi-step engineering equations in both metric and English conventions



Scan for my project portfolio

Water Billing System

Tech Stack: FastAPI, Granian, SQLAlchemy, MySQL, Docker, React Native, NFC, Xendit, Caddy, Cloudflare Tunnel

- Designed a utility billing platform that keeps confidential customer and payment data isolated on internal Docker networks, with role-scoped API access so each user reaches only what they need
- Built an offline-first meter reading app with tamper-proof NFC tags whose security keys derive on-device, so tag data never crosses the network
- Automated billing from a 5-tier progressive tariff and reconciled 22 payment channels end-to-end

GateKeeper

Tech Stack: Flask, itsdangerous, SQLite, Docker, Cloudflare Tunnel

- Single sign-on auth gate protecting a family of web apps with signed, tamper-evident cookies and forward-auth session verification
- Admin panel to issue and revoke access codes, with a backup code so the owner is never locked out

SolveSpace

Tech Stack: Flask, MySQL, Docker, Bubblewrap, Cloudflare Tunnel

- Practice sandbox that runs untrusted user code in isolated containers with hard memory, CPU, and process limits and no network access

TECHNICAL SKILLS

Languages: Python, C++, JavaScript, TypeScript, SQL

Backend: FastAPI, Flask, Granian, SQLAlchemy, Jinja2

Infrastructure: Docker, Caddy, Cloudflare Tunnel, Linux

Databases: MySQL, SQLite

Mobile: React Native, Expo

Hardware: Arduino, ESP32, Raspberry Pi, NFC (NTAG215)

Mechanical: AutoCAD, OnShape, Fusion 360